Integrated Platform for Dynamic GNN ETA Predict

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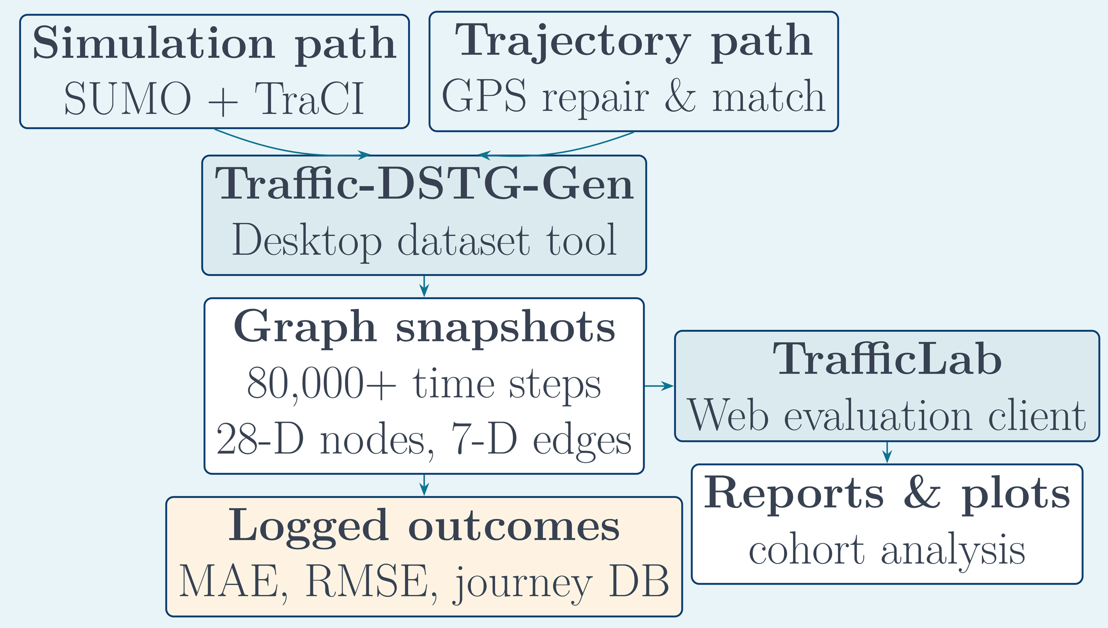
Motivation

Accurate **Estimated Time of Arrival (ETA)** underpins navigation, dispatch, and traffic management. **Dynamic Graph Neural Networks (GNNs)** model evolving road networks and vehicle interactions, but progress is limited by:

- Static-sensor datasets with few features
- Trajectory data lacking explicit route structure
- No unified path from *data creation* to *live model testing*

We present an **integrated open-source platform** that closes this gap: generate route-aware dynamic spatio-temporal graphs from SUMO simulation *or* real GPS trajectories, then evaluate trained ETA models interactively in a web environment.

End-to-End Workflow



Both construction paths export the **same graph schema**, enabling one training pipeline and one evaluation client.

Platform at a Glance

Traffic-DSTG-Gen

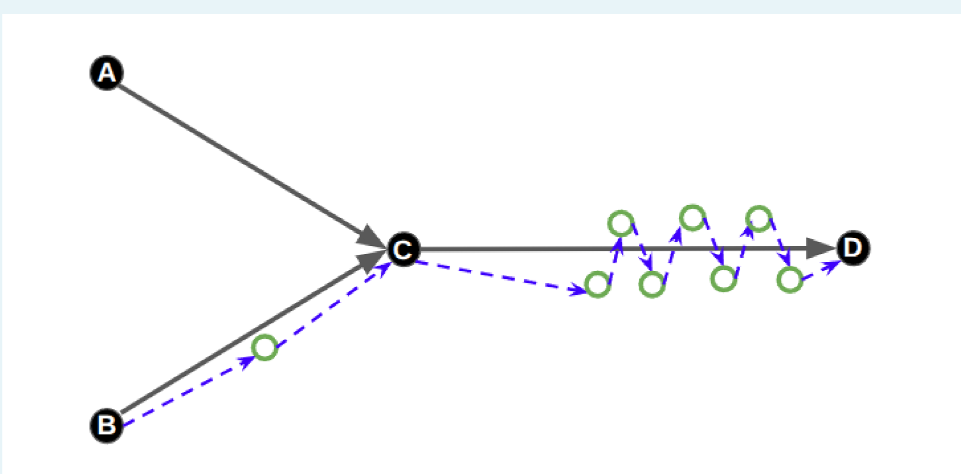
- Route-aware dynamic graphs from SUMO
- Hybrid static/dynamic nodes & edges
- Simulation & trajectory conversion
- Configurable temporal snapshots
- PySide6 desktop GUI

TrafficLab

- Real-time ETA inference
- Interactive trip evaluation
- Vue.js + FastAPI + PostgreSQL
- MAE / RMSE benchmarking
- Path-independent validation

Dataset scale	80,000+ snapshots
Node features	28 dimensions
Edge features	7 dimensions
Example bundles	3-zone city + Porto taxi

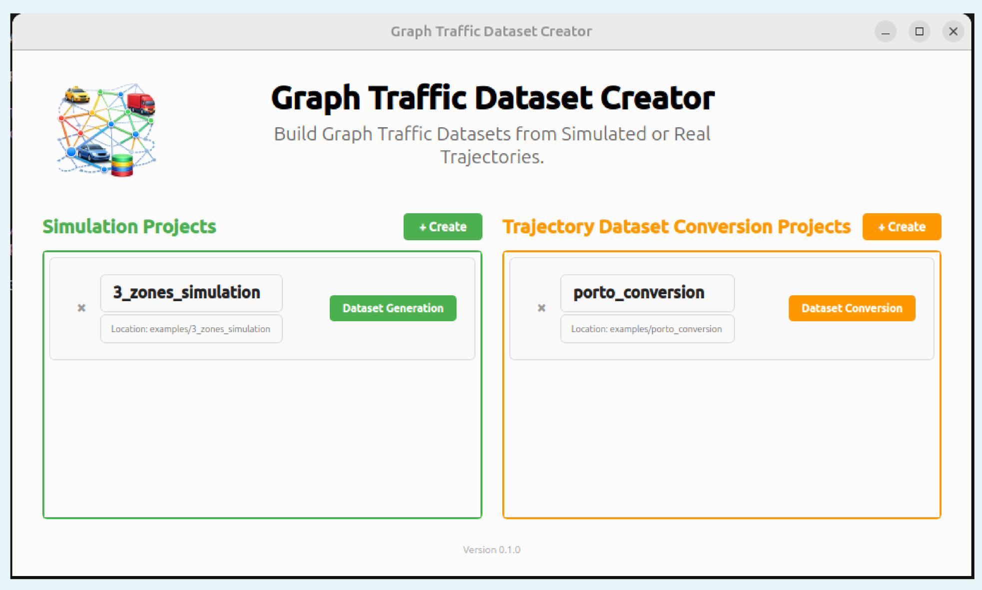
Route-Aware Dynamic Graph



Static layer: junction nodes (A–D) connected by road edges.
Dynamic layer: vehicle nodes with dashed relations to junctions and other vehicles, updated each snapshot.

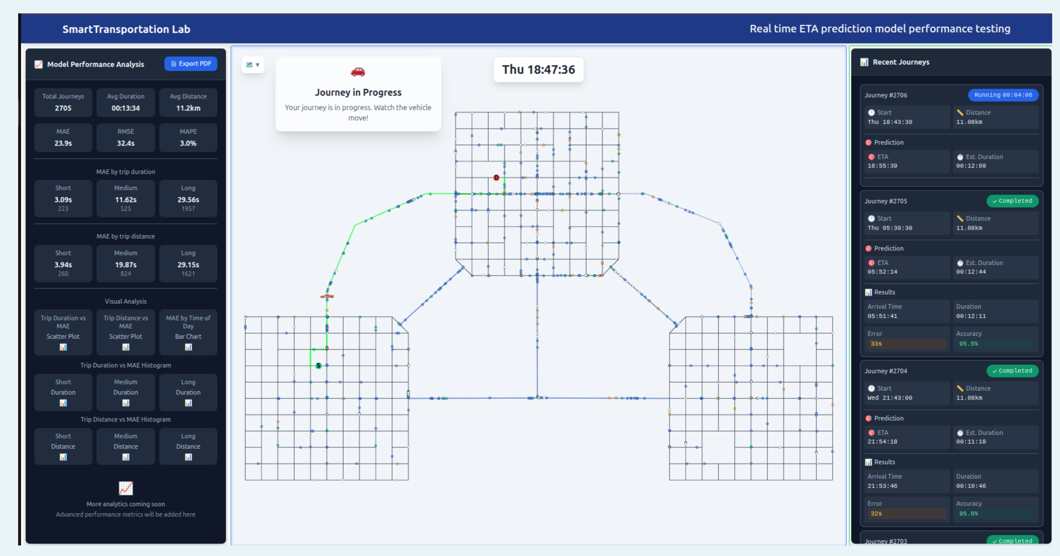
Features: speed, position, route progress, destination — enabling vehicle-level ETA prediction.

Dataset Generation (Desktop)



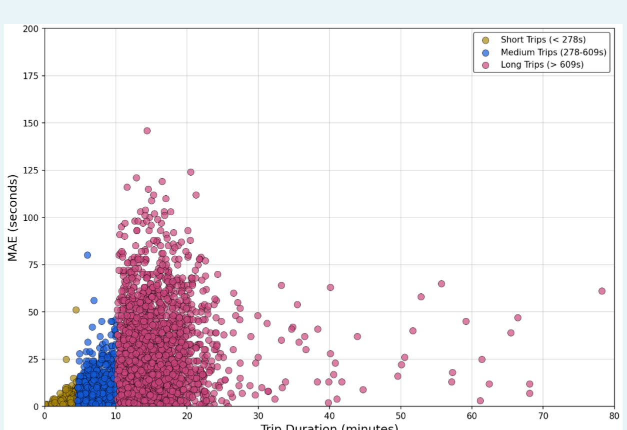
Two bundled examples: **3-zone city simulation** (left) and **Porto taxi conversion** (right). Operators configure SUMO scenarios or import GPS CSVs, validate routes on the map, and export time-stamped graph bundles for training.

Interactive Evaluation (Web)



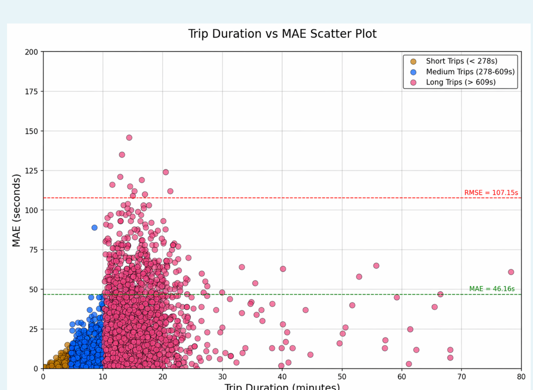
Users select origin/destination, receive a **predicted ETA**, run the journey in SUMO, and log realized duration. Dashboard panels show recent journeys and aggregate model performance — the same UI works for simulation-derived and trajectory-converted networks.

Evaluation Results



MAE vs. trip duration (ICSEE paper figure).

Short trips show lower error; the platform supports cohort analysis across duration buckets.

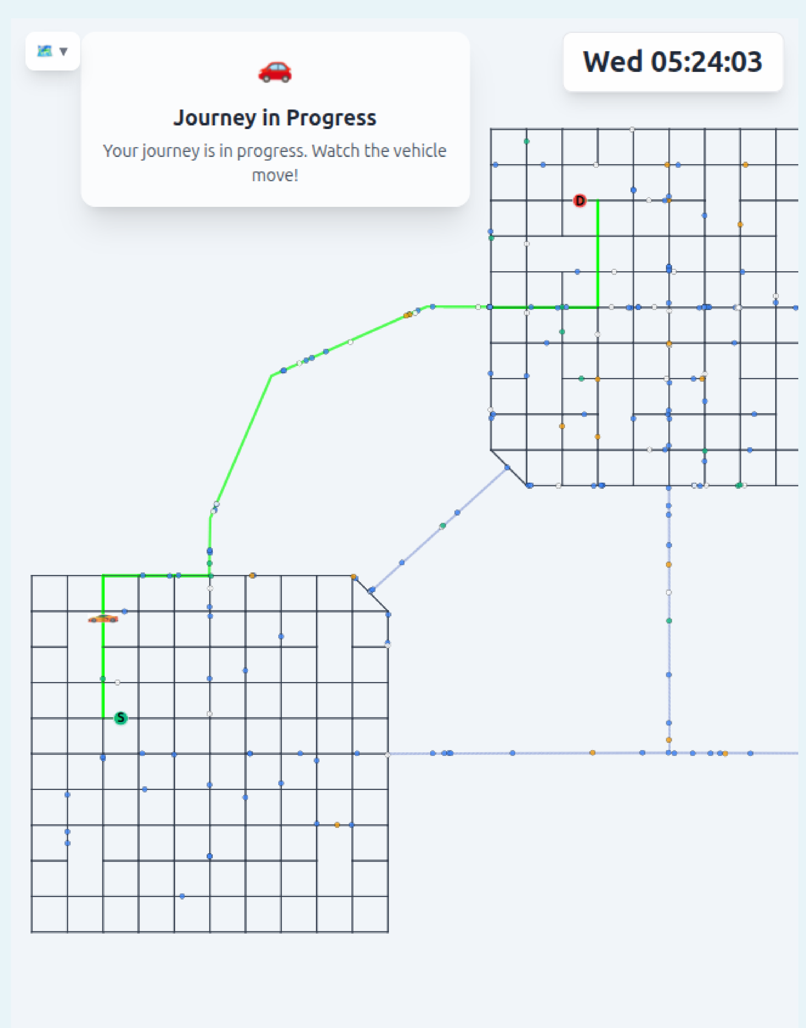


Per-journey absolute error vs. duration on the simulation bundle (2,707 logged trips).

	Sim.	**Porto**
Journeys (n)	2,707	3,055
MAE (s)	46.2	52.4
RMSE (s)	107.2	121.5
MedAE (s)	31.2	35.5

Web-tool metrics closely match held-out training evaluation, confirming end-to-end integration.

Example Journey



Journey #2709: 13.5 km route, predicted duration 17:11, realized arrival within **19 seconds** (98.1% accuracy). Each completed trip is persisted for aggregate reporting.

Conclusions & Links

Contributions

- Unified toolchain: SUMO simulation *and* GPS trajectory → dynamic graphs
- Route-aware representation suited to vehicle-level ETA GNNs
- Closed validation loop from dataset export to interactive web testing
- Open-source release for reproducible smart-transportation research

Future work: federated learning, additional real-world data sources, broader city deployments.

DSTG-Gen | TrafficLab | Live demo

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LaTeX TikZposter